A Supplementary

This supplementary provides additional implementation details, experiments, ablations, runtime and memory measurements, and qualitative comparisons to complement the main paper.

A.1 Additional Implementation Details

A.1.1 Training Data Sampling. Training data is drawn from a mixture of synthetic and real-world datasets containing clean, synthetically corrupted, and naturally in-the-wild image collections. At each iteration, we first sample a dataset and scene, and then sample between 2 and 16 views from that scene. The dynamic view sampler exposes the model to both sparse and denser multi-view configurations. Input images are resized such that their maximum spatial resolution is 448×448 pixels. Because the constituent datasets differ substantially in size and capture characteristics, the sampling procedure is designed to maintain diversity across indoor, outdoor, object-centric, and large-scale scenes.

A.1.2 Hybrid Supervision. The mask head predicts a static probability $M_i(p) \in [0, 1]$, where larger values indicate that pixel p is consistent with the persistent scene. The head is supervised using three complementary data sources:

- **Synthetic augmentation.** Transient objects are composited into clean training images using segmented foreground objects. The resulting binary transient masks are inverted to provide binary static-probability targets.
- **Pseudo-clean supervision.** For real in-the-wild image collections, WildGaussians [Kulhanek et al. 2024] is optimized offline and used to render transient-suppressed pseudo-clean observations. Its uncertainty maps are normalized to $[0, 1]$ and converted into soft static-probability targets:

$$Y_i(p) = 1 - U_i(p), \tag{14}$$

  where $U_i(p)$ denotes the normalized uncertainty at pixel p.
- **Real paired captures.** Real clean/in-the-wild pairs provide photometric and geometric supervision. When no mask annotation is available, these samples do not contribute to the static-probability binary cross-entropy loss.

Consequently, the mask loss is evaluated only for samples with binary synthetic targets or soft pseudo targets. This avoids treating unannotated pixels as static and prevents incorrect mask supervision on real paired data. Table 2 summarizes the supervision strategy.

A.1.3 Pseudo-clean Data Generation. Pseudo-clean supervision is generated once, offline, before student training. For each selected real-world scene, WildGaussians is optimized using the available in-the-wild images. The optimized representation is then rendered at the original camera viewpoints to produce transient-suppressed pseudo-clean images. The corresponding uncertainty maps provide soft targets for the student's predicted static probabilities. We generate pseudo-clean data for hold-out scenes from the Phototourism [Snavely 2006] and Dynamic RE10K [Chen et al. 2026] datasets.

This preprocessing stage is not part of the inference pipeline. At test time, WildAnySplat processes the input images directly with the student network and does not require WildGaussians, pseudo-clean rendering, or scene-specific optimization.

A.1.4 Optimization Details. The model is trained for 35 000 iterations using AdamW and a cosine learning rate schedule. The peak learning rate is 2×10^{-5} with a warm-up of 2 000 iterations. We accumulate gradients over four consecutive mini-batches before each optimizer update. This increases the effective batch size while remaining within the memory constraints of transformer-based, multi-view Gaussian prediction.

The loss weights are $\lambda_{\text{photo}} = 1.0$, $\lambda_{\text{distill}} = 0.1$, and $\lambda_{\text{bg}} = 10^{-4}$. The bilateral-grid resolution is $16 \times 16 \times 8$. The clean-image teacher is initialized from pretrained VGGT weights and remains frozen throughout training.

Table 2. Supervision strategy for the different training data sources. A dash indicates that no mask supervision is applied.

Source	Clean supervision	Mask supervision
Synthetic augmentation	GT	Binary
Pseudo-clean data	Pseudo	Soft
Real paired captures	GT	–

Table 3. Inference runtime (seconds) and peak GPU memory (GB) measured on a single GPU with 140 GB VRAM.

Runtime (s)

	Number of Input Views					
Method	3	6	9	12	24	48
AnySplat	2.76	3.13	3.53	3.79	4.78	5.91
WildAnySplat	4.29	4.61	5.09	5.37	6.62	8.34

Peak GPU Memory (GB)

	Number of Input Views					
Method	3	6	9	12	24	48
AnySplat	3.96	5.82	7.64	9.52	13.64	18.81
WildAnySplat	6.18	8.03	10.13	11.81	15.83	20.55

A.2 Runtime, Memory, and Preprocessing Cost

Table 3 reports end-to-end inference time and peak GPU memory for AnySplat and WildAnySplat. Measurements were obtained on a single GPU with 140 GB of memory. Runtime includes feed-forward prediction of the scene representation; WildAnySplat additionally predicts transient probabilities and bilateral grids.

The additional cost primarily arises from the transient-mask prediction and bilateral-grid head. The growth in memory remains moderate as the number of input views increases: at 48 views, peak memory increases from 18.81 GB for AnySplat to 20.55 GB for WildAnySplat.

Data preprocessing to generate pseudo-clean data involves running per-scene optimization of WildGaussians on multiple hold-out training scenes independently. Each scene-specific training run takes roughly 4 hours on a single GPU with 32 GB of memory. We

Table 4. Component ablation on NeRF On-the-Go. BG denotes bilateral-grid appearance modeling and TM denotes transient-mask prediction.

Variant	PSNR ↑
AnySplat baseline	14.47
Baseline + BG	15.65
Baseline + TM	15.75
Baseline + TM + BG (WildAnySplat)	**16.12**

Table 5. Effect of view overlap on RealX3D. Lower sampling stride produces greater overlap between context views, allowing transient reasoning to operate more effectively.

Sampling Stride	AnySplat	WildAnySplat	Gain
Stride 8	11.12	**11.97**	+0.85
Stride 4	14.41	**15.56**	+1.15
Stride 1	19.62	**21.66**	+2.04

parallelize this process across multiple GPU nodes to speed up data generation time.

A.3 Component Ablation

We evaluate the contributions of bilateral-grid appearance modeling (BG) and transient-mask prediction (TM) on NeRF On-the-Go [Ren et al. 2024]. Table 4 reports PSNR for the AnySplat baseline and for models augmented with each component independently and jointly.

Both components independently improve the baseline. Bilateral grids increase PSNR by 1.18 dB, showing that explicitly accounting for image-dependent appearance variation improves the recovery of canonical scene colors. Transient masks increase PSNR by 1.28 dB, demonstrating the benefit of suppressing unreliable observations during supervision and Gaussian fusion. Combining both components produces the best result, improving the baseline by 1.65 dB. The combined gain indicates that transient suppression and appearance normalization address complementary sources of multi-view inconsistency.

A.4 Effect of View Overlap

The transient reasoning of our model relies on repeated observations of persistent scene content. To study this effect, we evaluate WildAnySplat on RealX3D [Liu et al. 2025a] while varying the temporal sampling stride between context views. A smaller stride selects nearer frames and therefore generally increases image overlap. The number of input views is kept fixed.

Performance improves substantially as context views become more overlapping: reducing the stride from 8 to 4 improves PSNR by 3.59 dB, while reducing it from 4 to 1 yields a further 6.10 dB gain. These results highlight that sparse sampling and limited co-visibility constitute a particularly difficult regime for feed-forward reconstruction.

The result is also consistent with the mechanism of our method. A transient object can be identified reliably when the underlying static region is visible in other views. When an occluder appears in the only observation of a region, the evidence available to distinguish persistent and transient content is fundamentally limited. Increased overlap provides more cross-view evidence for both geometric reconstruction and transient suppression.

A.5 Transient Masks and Semantic Segmentation

Our mask head predicts the probability $M_i(p)$ that an observation belongs to the persistent static scene. For visualization, we invert this prediction to obtain the transient probability

$$T_i(p) = 1 - M_i(p), \tag{15}$$

where brighter pixels indicate regions that are more likely to be transient.

Unlike semantic segmentation masks, these predictions do not represent predefined object categories. Semantic identity alone does not determine whether an observation should be included in the reconstruction. For example, a parked vehicle that remains fixed throughout a capture constitutes persistent scene content, whereas a moving pedestrian, temporary banner, or construction object may be transient.

Methods based on semantic masks, such as GenWildSplat [Gupta et al. 2026], may therefore remove persistent objects that belong to nominally dynamic categories while failing to suppress inconsistent objects outside those categories. WildAnySplat instead learns to identify transient observations using multi-view consistency signals provided by synthetic masks, pseudo uncertainty, clean-teacher predictions, and reconstruction losses. The predicted static probabilities are used to downweight unreliable photometric supervision and suppress inconsistent per-view Gaussians during voxel fusion.

Figures 9–13 provide additional qualitative comparisons of the resulting transient probabilities. WildAnySplat identifies view-specific inconsistencies using cross-view evidence without relying on predefined semantic object categories.

A.6 Relationship Between Qualitative and Quantitative Results

The results of the main paper show visually substantial reductions in ghosting or duplicated transient objects, but yield modest changes in PSNR, SSIM, or LPIPS. There are three main reasons for this:

Sensitivity to pose and image alignment. Pixel-based metrics penalize small rendering-to-ground-truth misalignments. Sparse, unposed, in-the-wild inputs make camera estimation challenging, and a minor pose error can lower PSNR and SSIM substantially across large image regions even when the reconstructed geometry or transient handling is perceptually improved.

Limited overlap in sparse-view protocols. Transient removal requires evidence that separates persistent scene content from view-specific occlusion. When the context images have little overlap, some regions are observed only once. In this setting, suppressing a transient object without hallucinating the hidden background is intrinsically ambiguous. The overlap experiment in Table 5 demonstrates the strong dependence of reconstruction quality on available cross-view evidence.

Metric dilution. Transient artifacts may occupy a relatively small fraction of the evaluation image. Removing a transient object such as

a pedestrian-shaped ghost can produce a large perceptual improvement while changing only a small fraction of all evaluated pixels. Full-image average metrics can therefore understate improvements in the regions most affected by in-the-wild inconsistencies.

For these reasons, the quantitative results should be interpreted together with the qualitative comparisons. The metrics measure overall image fidelity, whereas the visual results more directly expose transient duplication, geometric ghosting, and cross-view color inconsistency.

A.7 Failure Cases and Limitations

WildAnySplat remains limited when dynamic or unreliable regions dominate the input images. If most views contain large moving objects and reveal little persistent background, camera estimation and geometric reconstruction can both degrade. The predicted masks cannot recover static content that is never observed.

The bilateral-grid model is designed primarily for smooth, spatially varying photometric transformations. It can compensate for exposure, white balance, contrast, vignetting, and moderate local color shifts, but it may not fully model severe illumination changes, sharp cast shadows, strong view-dependent effects, or highly nonlinear camera processing. Under such conditions, residual color differences may remain in the canonical reconstruction.

Pseudo-clean supervision can also inherit errors from the optimization-based method used to generate it. Although uncertainty weighting reduces the impact of unreliable regions, systematic artifacts in pseudo-clean renders may still bias the student. Finally, WildAnySplat reconstructs a canonical static scene; it does not explicitly model object motion or produce a temporally varying dynamic representation.

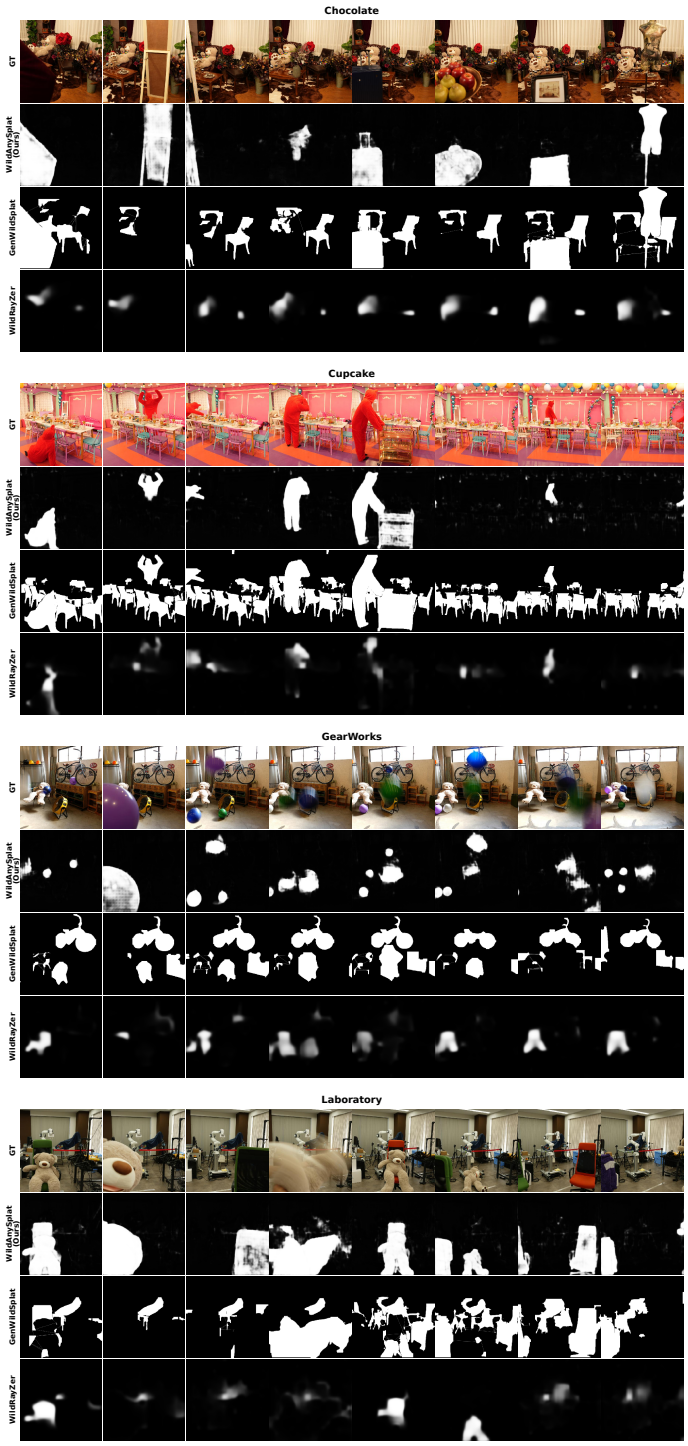

Fig. 9. Qualitative transient-mask comparison on RealX3D. The top row of each scene shows input RGB images, followed by transient probabilities predicted by WildAnySplat, GenWildSplat, and WildRayZer. WildAnySplat more consistently identifies view-dependent occluders and retains persistent scene content.

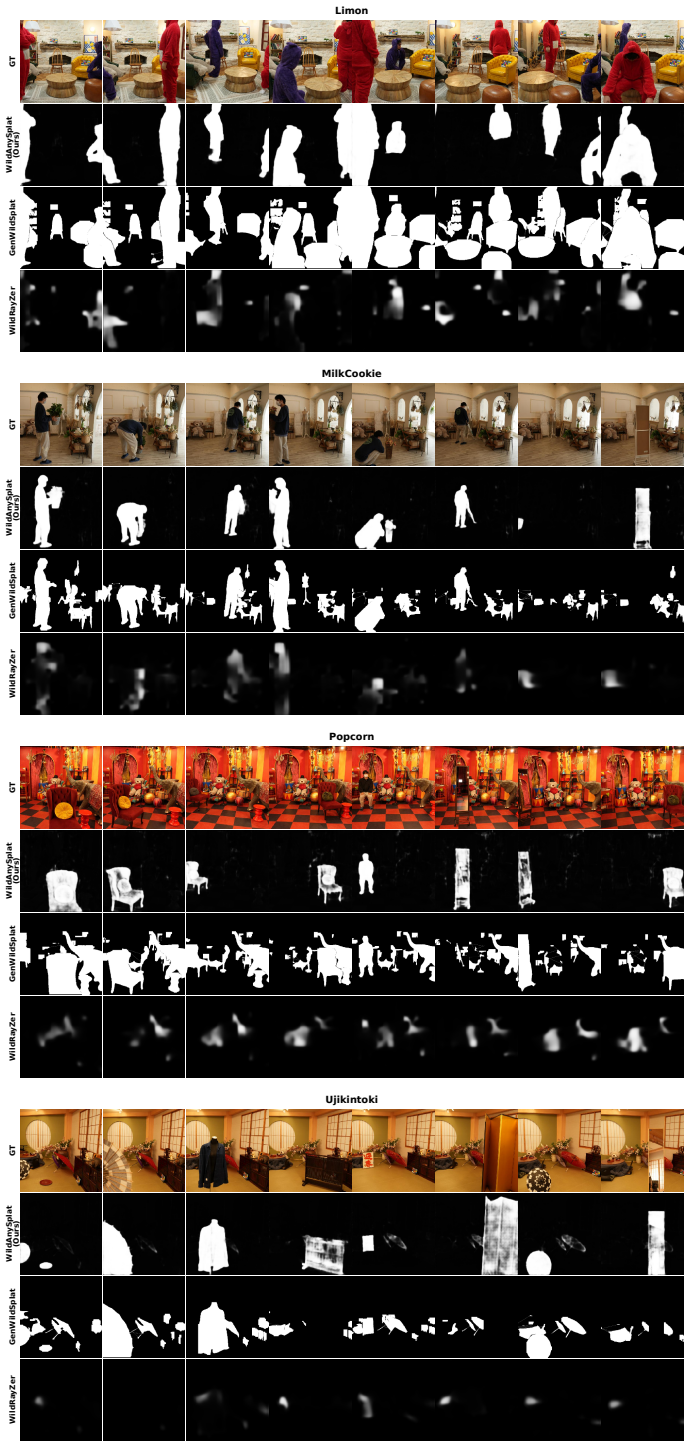

Fig. 10. Qualitative transient-mask comparison on RealX3D. The top row of each scene shows input RGB images, followed by transient probabilities predicted by WildAnySplat, GenWildSplat, and WildRayZer. WildAnySplat more consistently identifies view-dependent occluders and retains persistent scene content.

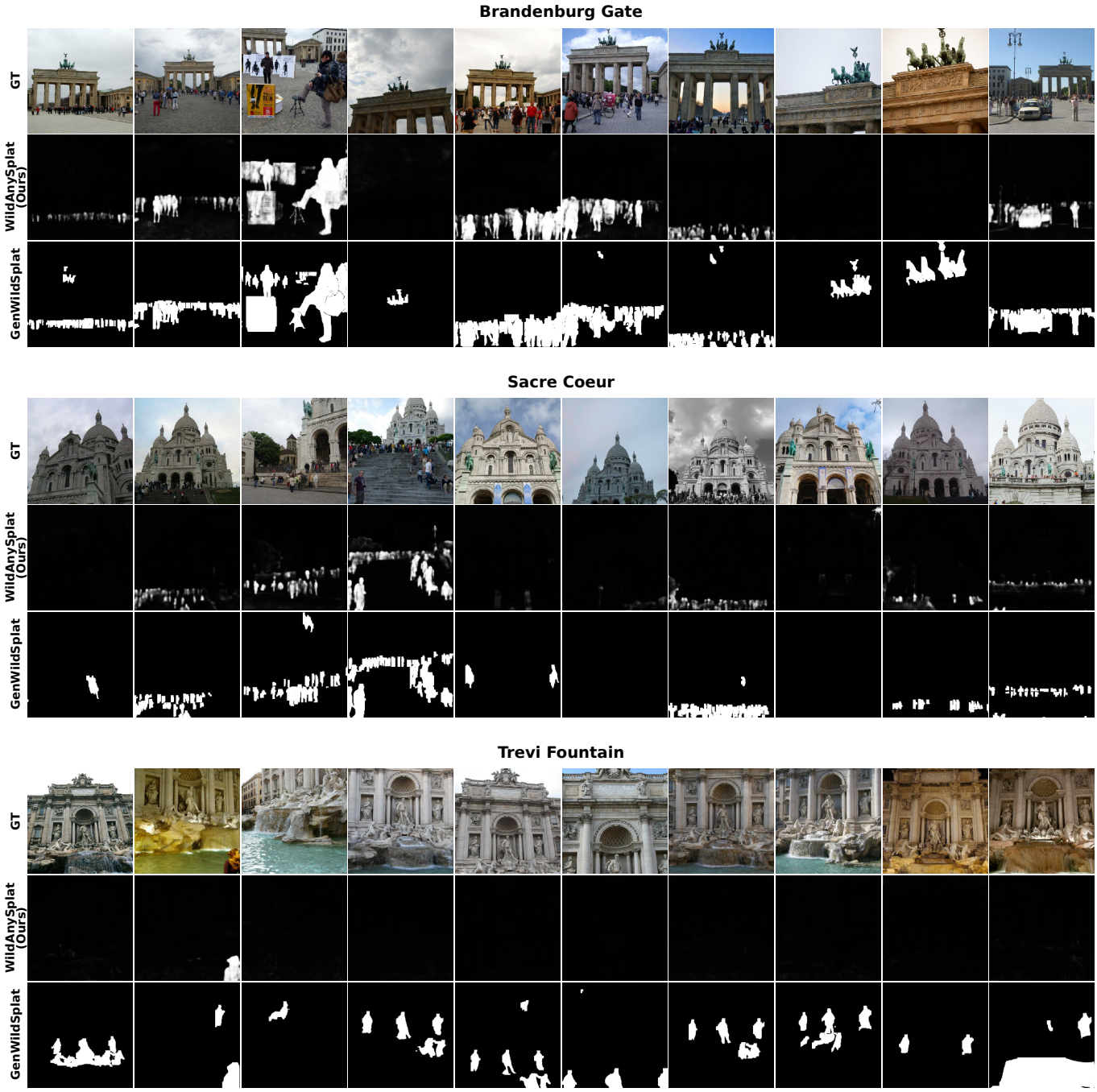

Fig. 11. Qualitative transient-mask comparison on Phototourism. For each scene, the top row shows input RGB images, followed by transient probabilities predicted by WildAnySplat and GenWildSplat. The examples contain tourists, vehicles, and other objects that are inconsistent across the image collections.

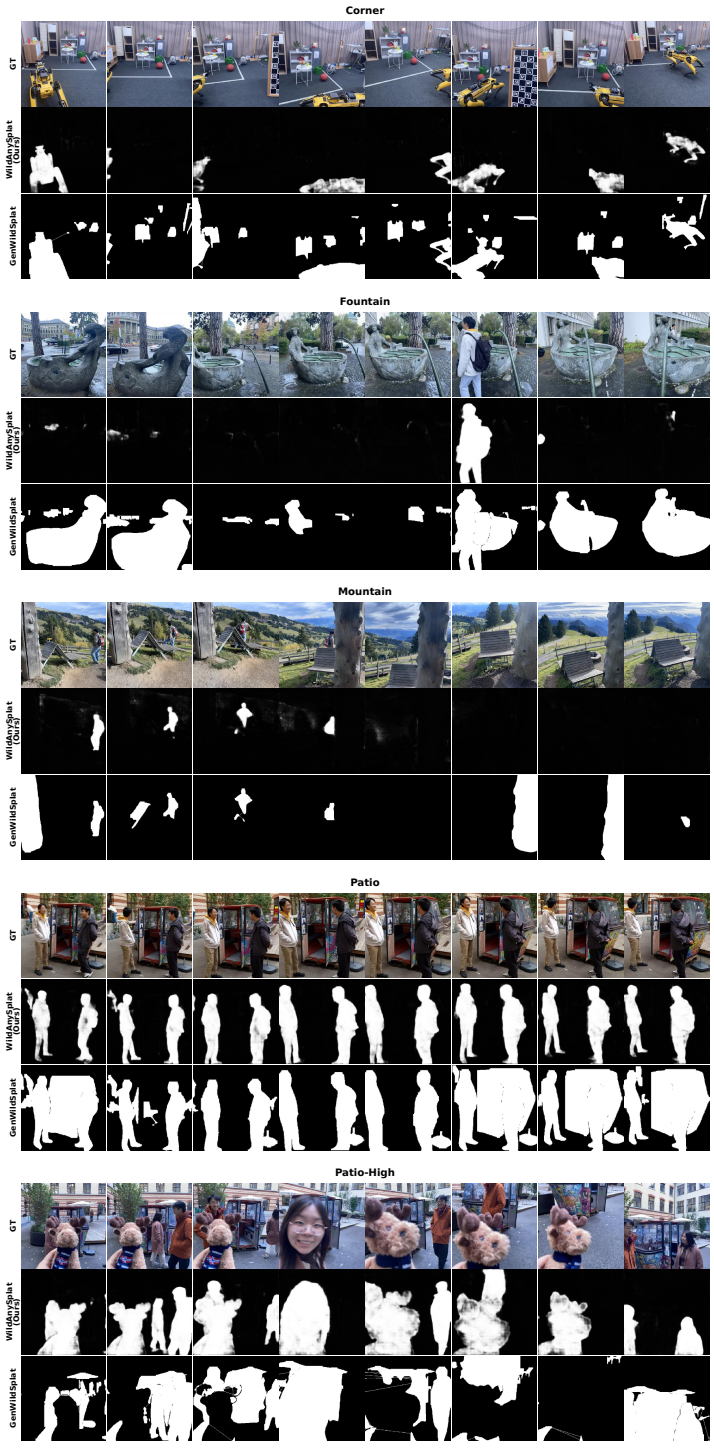

Fig. 12. Qualitative transient-mask comparison on NeRF On-the-Go. The top row of each scene shows input RGB images, followed by transient probabilities predicted by WildAnySplat and GenWildSplat. WildAnySplat better detects actual transient objects while ignoring persistent static content (e.g. the fountain).

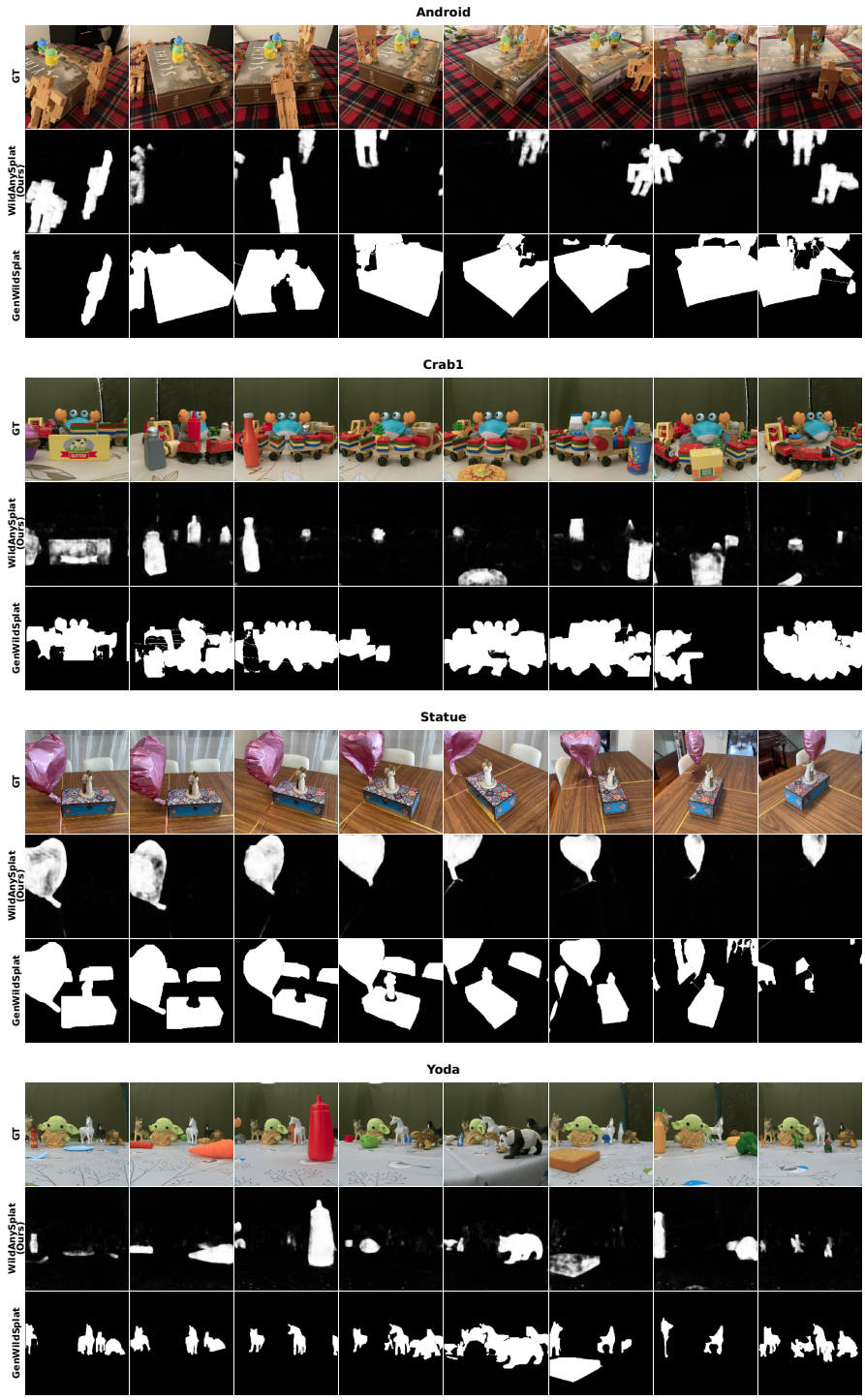

Fig. 13. Qualitative transient-mask comparison on RobustNeRF. The top row of each scene shows input RGB images, followed by transient probabilities predicted by WildAnySplat and GenWildSplat. WildAnySplat suppresses view-specific clutter and preserves regions that remain consistent across the capture.